

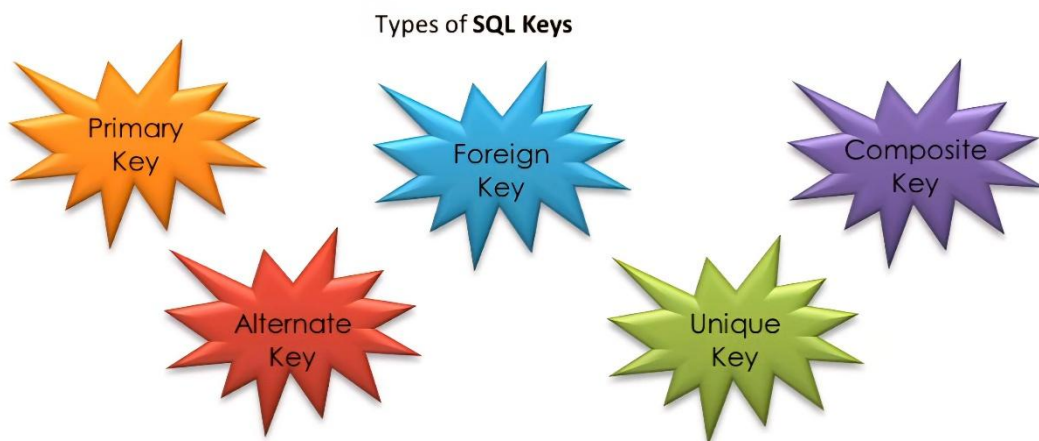
SQL (Structured Query Language)

➤ Keys in SQL :-

- Keys are special fields or a combination of fields in a database table.
- They are used to uniquely identify each record in the table.
- Keys help in linking two or more tables together.
- They ensure data integrity and avoid duplication of records.

➤ Types of SQL Keys :-

Different types of keys are used for different purposes, such as uniqueness, relationship creation, and data validation.



1) Primary Key :-

Primary Key

- **pk-** that uniquely identifies each row in the table.
- Primary key always has unique data.
- A primary key cannot have null value.

Syntax

```
create table table_name(column_name datatype(size),  
PRIMARY KEY(column_name));
```

Eg. **create table** student(id int, name varchar(20), **PRIMARY KEY**(id));

CustomerID	CustomerName	PhoneNumber
1	Rohit	9876543210
2	Sonal	9765434567
3	Ajay	9765234562
4	Aishwarya	9876567899
5	Akash	9876541236

2) Foreign Key :-

Foreign Key

- Establish a link between two tables.
- Foreign key in one table used to point primary key in another table.

Syntax

create table second_table_name(column_name datatype(size),
PRIMARY KEY(column_name),**FOREIGN KEY**(column_name)
REFERENCES first_table_name(first_table_column_name));

Eg. **create table** student2(s_id int, primary key(s_id), foreign
key(s_id) references student1 (id));

Primary Key

Foreign Key

ID	Name	Course
2041	Tom	Java
2204	John	C++
2043	Alice	Python
2032	Bob	Oracle

Student Details

ID	Marks
2041	65
2204	55
2043	73
2032	62

Student Marks

3) Composite Key :-

Composite Key

- A primary key that is made by the combination of more than one attribute is known as a composite key.

Syntax

```
create table table_name(column_name1 datatype(size),  
column_name2 datatype(size), PRIMARY KEY(column_name1,  
column_name2));
```

Eg. **create table** student(id **int**, name **varchar**(20), **PRIMARY KEY**(id,name));

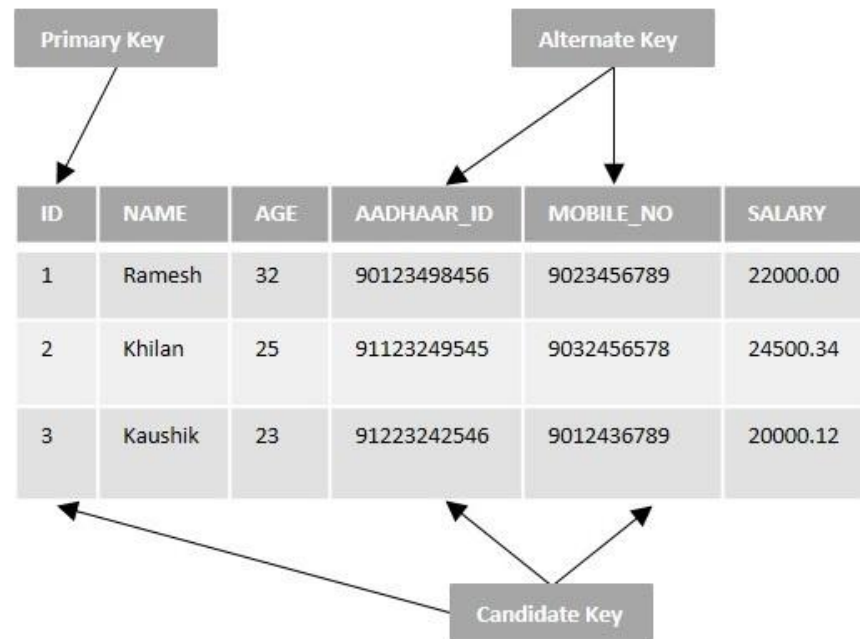
Composite Key					
ID	NAME	AGE	AADHAAR_ID	MOBILE_NO	SALARY
1	Ramesh	32	90123498456	9023456789	22000.00
2	Khilan	25	91123249545	9032456578	24500.34

4) Alternate Key :-

Alternate Key

- Each row of first table is combined with each row from the second table, known as Cartesian join or cross join.
- Eg. Id, Roll_no, Email are qualified to become a primary key. But since StudID is the primary key, Roll No, Email becomes the alternative key.

Id	Roll_no	Email
101	1	abc@gmail.com



5) Unique Key :-

Unique Key

- set of one or more than one fields/columns of a table that uniquely identify a record in a database table.
- it is little like primary key but it can accept only one null value.
- it cannot have duplicate values.
- There may be many unique key constraints for one table, but only one PRIMARY KEY constraint for one table.

Syntax

```
create table table_name(column_name1 datatype(size),
column_name2 datatype(size), UNIQUE(column_name1));
```

Eg. **create table** student(id int NOT NULL, name varchar(20), **UNIQUE**(id));

Name	Age	SSN
Amy Smith	24	123123123123
Jim Anderson	21	456456456456
George Brown	32	456456456456

This is illegal because the SSN column implements the UNIQUE constraint.

Name	Age	SSN
Amy Smith	24	123123123123
Jim Anderson	21	456456456456
George Brown	32	789789789789

This is correct.

➤ Difference Between Primary & Foreign Key :-



- primary key cannot be null
- Primary key is always unique
- Primary key uniquely identify a record in the table.
- There is only one primary key in the table

- Foreign key can be null
- Foreign key can be duplicate
- Foreign key is a field in a table that is primary key in another table.
- We can have more than one foreign key in the table.